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Cultivating research skills through scholarly digital storytelling

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ABSTRACT

A fundamental responsibility of higher education institutions, across disciplines, is to develop student research skills. The exponential growth of digital scholarship, however, challenges traditional definitions of research. There is a need for deeper understanding of the ways in which students can develop and implement academic research skills and digital research skills simultaneously. This qualitative study explored graduate student engagement in scholarly digital storytelling, a technology-enhanced assessment, across multiple disciplines at a research university in the United States. Findings suggest that students developed the essential research skills of autonomy and flexible thinking along with multimodal research skills throughout the semester-long process of creating scholarly digital stories. Implications for utilizing technology-enhanced assessments, such as scholarly digital storytelling, in higher education are discussed.

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Introduction

Institutions of higher education have increasingly focused on student research skills as a key learning outcome (Willison, 2018). At the same time, there is a pressing need to integrate digital research skills into higher education and to rethink scholarship in the digital age (Literat et al., 2018). These are not mutually exclusive and complement the shift in student learning from reproduction of knowledge to knowledge production (Literat et al., 2018; Passey et al., 2018; van Laar et al., 2017). Scholarly digital storytelling, a technology-enhanced assessment, requires students to combine research and digital skills to communicate academic work to an intended audience (Schrum, 2021; Snelson, 2018). While research on technology-enhanced assessments has focused on increasing digital literacy and information literacy (Spante et al., 2018), there has been less focus on student learning and development of research skills (Oliver & Jorre de St Jorre, 2018; Price & Kirkwood, 2014). This article presents a qualitative study of graduate students who engaged in scholarly digital storytelling at a research-intensive university in the United States over the course of one semester and asks the following research question: Does scholarly digital storytelling promote student development of traditional and digital research skills?

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Literature review

Systematic efforts to define core research skills include the Research Skill Development (RSD) framework with a focus on facilitating student learning 'in ways that enable their skills to grow in sophistication and rigour' (Willison, 2018, p. i; Willison & O'Regan, 2007). Research is often defined as difficult and challenging, including tasks such as defining strong research questions, collecting and analyzing data to address those questions, drawing conclusions, and sharing results (Kiley & Wisker, 2009; Rodríguez et al., 2019). Academic research requires flexible thinking and persistence (van Laar et al., 2017; Willison, 2018) and these skills must be taught explicitly (Willison, 2012). This overlaps in interesting ways with digital literacy skills which include the ability to independently 'research, organize and synthesize information through digital technology' (Sparks et al., 2016, p. 3) and are best acquired through iterative development (Burdick et al., 2012; Snelson, 2018). Students use technology regularly in their daily lives, but this ability does not automatically transfer to facility with academic digital skills (Henderson et al., 2017; Selwyn, 2016).

Assessments that emphasize research skills and integrate technology to transform learning require planning, scaffolding, and support (Ng'ambi & Bozalek, 2015). They typically focus on inquiry-based learning, engaging learners as knowledge producers rather than as knowledge consumers (Passey et al., 2018; van Laar et al., 2017). Literat et al. (2018) expand on this concept by advocating for the flexibility to 'embrace whatever mode of inquiry, analysis, or representation' (p. 567) is relevant to produce and communicate quality scholarly digital content. Creating authentic academic digital projects expands traditional research skills as meaning is created through design as well as multimedia editing and development (Burdick et al., 2012; Literat et al., 2018) with the goal of empowering students to evaluate and produce digital content in order to communicate effectively (Literat et al., 2018; Willison, 2018). Higher education assessments have not kept pace in helping students to develop these skills (Bond et al., 2018; Castañeda & Selwyn, 2018; Englund et al., 2017; Mercader & Gairín, 2020; Silber-Varod et al., 2019).

The RSD framework emphasizes 'research mindedness' with a focus on curiosity as students generate questions, locate and evaluate sources, organize and analyze information, and communicate new knowledge (Willison, 2018; Willison & O'Regan, 2007). Scholarly digital storytelling builds on this framework and incorporates digital literacy to cultivate research skills for the digital age, including autonomy, flexible thinking, and multimodal research. Autonomy is a key factor throughout all facets of the RSD framework, ranging from prescribed and bounded research to open-ended and unbounded research (Willison, 2018). Assessments that involve high levels of autonomy as defined in the RSD framework include student determination of 'what is investigated and how that is done' (Willison & O'Regan, 2007, p. 400). Other researchers similarly emphasize self-awareness of and responsibility for one's own learning (Gyuris, 2018) or self-directed learning (Carvalho, 2016) as a critical part of the research process, including digital autonomy (Passey et al., 2018).

Academic research also requires flexible thinking, including persistence and iterative development (van Laar et al., 2017; Willison, 2018). In the RSD framework, students learn not only to collect and evaluate information based on self-generated criteria but

to engage in problem solving, to 'reflect' and 'renew' their processes and thinking, analyze and synthesize information with a focus on 'incremental and cyclic development,' and respond to feedback and new ideas (Willison, 2018, p. 2). Carvalho (2016) discusses the importance of flexible thinking and flexible knowledge as key transferable skills. Flexible thinking, including iterative thinking and development, is also central in research on digital skills and includes approaching conceptual and technical problems with a 'trial and error attitude' (Iordache et al., 2017, p. 24). Flexible thinking requires a valuing of process as well as product (Burdick et al., 2012) which is an essential part of teaching and learning with scholarly digital storytelling (Schrum, 2021).

While the RSD does not focus specifically on digital skills, there are central components that adapt readily to digital, multimodal research grounded in academic practice, including identifying a range of sources, integrating various components of research to create new knowledge, and presenting that knowledge in ways that are appropriate for the intended audience (Willison, 2018). Multimodal research skills are further emphasized in emerging frameworks for digital competence and digital research skills that focus on analyzing different forms of digital content (Carretero et al., 2019; Redecker & Punie, 2017) and producing and sharing knowledge in new formats, including digital storytelling (Literat et al., 2018). Learning how to analyze information across media is an essential research skill for the digital age (Passey et al., 2018; Snelson, 2018).

In a study about efforts to incorporate the RSD framework, Willison (2012) writes of the importance of 'explicit development of student research skills' (p. 905) which is similarly emphasized by Oliver and Jorre de St Jorre (2018) in their study of graduate attributes for 2020 and beyond. Autonomy, flexible thinking, and multimodal research skills are often novel for students and require teachers to explicitly facilitate skill development in a scaffolded and cyclical way. Scholarly digital storytelling is designed to purposefully combine skill development in these areas with a focus on creating digital content for an intended audience (Schrum, 2021). Technical skills, such as filming and editing are taught in class and developed alongside critical thinking skills, creativity, and disciplinary thinking (Peñalba et al., 2020; Snelson, 2018). The inquiry-based structure allows students to engage in discipline-specific research strategies while deepening their understanding of the iterative, reflective, and autonomous nature of research (Willison, 2012).

Methods

This article presents a qualitative study of graduate students from multiple disciplines who engaged in a semester-long scholarly digital storytelling course at a research university in the United States and asks the following research question: Does scholarly digital storytelling promote student development of traditional and digital research skills? Schrum has taught scholarly digital storytelling as a 15-week course six times since 2010 and observed significant growth in academic research skills alongside digital skills. In 2018, Schrum began to collect data to investigate this observation, including student work, such as blog posts, reflections, and digital stories, and subsequently conducted individual, semi-structured interviews with 32 graduate students, close to half of all students who have taken the course. All interviews were conducted after the course ended and final grades had been submitted to minimize influence of the

teacher-student relationship. Written informed consent was obtained from all participants and research was conducted in accordance with institutional ethical standards.

Price and Kirkwood (2014) call for more evidence of the 'impact on student learning' related to technology use in higher education classrooms (p. 556). The data in this study offers insight into immediate student experiences, including reflections and digital stories created during the course, as well as longer-term impact on research skill development as seen through student interviews conducted after the course ended. Among the 32 students interviewed, 22 identify as female, nine as male, and one as non-binary. All names are pseudonyms. The students who participated in this research came from the disciplines of history, higher education, and rhetoric and represented different phases of graduate education, from first-year master's through advanced doctoral (see Table 1). Each student individually created a 10-minute scholarly digital story based on their academic research. This was the major course assessment and required research, effective scholarly communication, and digital knowledge production as reflected in the syllabus and rubric (George Mason University, 2021).

Interview questions focused on student learning in the course, including skill development and future research goals. All interviews were recorded, transcribed verbatim, verified, and validated through member checking and, along with student course work, coded and analyzed using thematic analysis (Braun & Clarke, 2006; Nowell et al., 2017). Initial phases of familiarization and code generation were followed by theme development and review. During this iterative process, additional themes were developed and the data was reviewed again for referential adequacy (Nowell et al., 2017). Schrum and Bogdewiecz reviewed the data and the themes both independently and collaboratively, discussing similarities and discrepancies during theme development and before the final round of coding. Initial codes were broad and covered a range of topics. Upon further review of data and consultation between both authors, three final themes were derived that were broad enough to encapsulate the underlying sentiment in the data, but discrete enough so as to not overlap with one another. Results of this study will be presented under these final themes: autonomy, flexible thinking, and multimodal research. All student quotations come from student interviews unless otherwise indicated.

Table 1. Student degree levels and disciplines.

Student Name (pseudonym)	Degree Level	Discipline
Adam	Doctoral	History
Amanda	Master's	History
Ashley	Master's	History
Bailey	Doctoral	Rhetoric
Chad	Master's	History
Chelsea	Doctoral	Higher Education
Karen	Doctoral	Rhetoric
Ken	Doctoral	History
Lauren	Master's	Higher Education
Mark	Doctoral	Higher Education
Morgan	Master's	History
Roberto	Doctoral	History
Taylor	Master's	History
Tracy	Master's	History

Study context

Course objectives included creating a final project grounded in academic research and learning digital skills necessary to communicate scholarly work (George Mason University, 2021). Students defined topics related to their academic interests, an open inquiry approach that fits with Willison's definition of 'unbounded' research (2018, p. 2). While core academic research skills, such as identifying and analyzing sources, conducting interviews, and developing a strong thesis (Willison, 2018) can be found in many discipline-specific courses, this course also required students to engage in multimodal research and knowledge production by incorporating audio, video, and images into the scholarly argument (Literat et al., 2018; Passey et al., 2018).

Assignments were purposefully scaffolded to provide autonomy while helping students strengthen research, digital, and storytelling skills grounded in disciplinary thinking as the semester progressed. Skills included learning to think visually and to produce digital content via low-stakes assignments, such as telling a story with ten images, creating a two-minute video diary, and producing a 30-second promotional trailer. The final project was similarly broken down into discrete assignments to encourage iterative conceptual and technical development. Students, for example, submitted a project pitch during the second week of class, including the topic, approach, and research plan, and a revised project pitch the following week. In subsequent weeks, students conducted research, refined their topics, experimented with digital production, and tracked their progress through updates and assignments such as storyboards for mapping argument, content, and multimedia components. Students submitted an annotated bibliography documenting their engagement with relevant literature as well as a copyright assessment and a treatment (description of visuals, audio, and narration) by the eighth week of the semester. The rough cut (rough draft) was due by the eleventh week. The class then critiqued the revised digital stories in the remaining weeks, and the final version and reflection were due at the end of the semester (George Mason University, 2021). The course is tool agnostic by design in order to accommodate student preferences and budgets, meaning that some students use subscription-based software, such as Adobe Premiere, while others use freely available options, such as iMovie.

Projects reflected scholarly practice as defined by the student's discipline. Higher education students, for example, addressed a central question grounded in contemporary theory and practice, such as student development theory for online learners or career readiness among college student athletes. History students integrated primary and secondary sources to present an original argument about a historical topic, such as a museum renovation or the life and legacy of an African American entrepreneur. Content must be presented in an engaging and relevant way for the intended audience and technical requirements included smooth transitions and quality audio and visuals.

Students received timely and extensive formative feedback, essential for successful skill development (Gyuris, 2018; Willison, 2012), throughout the semester. The teacher responded to each assignment, including the initial and revised project pitches, storyboard, interview questions, release forms, annotated bibliography, treatment, rights assessment, credits, rough cut, and project updates. The teacher met individually with each student at least once during the semester to discuss individual research goals and process as well as overall progress. In addition, students provided ongoing peer

feedback on project pitches, storyboards, treatments, and rough cuts, sharing research processes and resources within and across disciplines.

Results

As students worked through course content and assignments and started to conceptualize, research, and create their own scholarly digital stories, they practiced and implemented a range of skills. The following section examines student development throughout the semester-long course through the themes of autonomy, flexible thinking, and multimodal research.

Autonomy

Students engaged in open inquiry (Willison et al., 2017; Willison & O'Regan, 2007) as they defined their topics, identified audiences, conducted research, crafted and refined arguments, and produced scholarly digital stories. The course design fostered these skills through scaffolded assignments requiring students to identify their own research agenda. Initial topic ideas were often broad or overly ambitious, but through written and in-person conversations with the teacher and classmates, students articulated the steps and processes for the final project which typically led to revision and clarification. This process involved learning by doing, including productive failures along the way, but the structure of the course and the ongoing feedback and engagement supported reframing and persistence.

For some, this level of autonomy was a new experience. Morgan explained that 'This was actually the first class where someone asked me what my research study, research topic is. And I wasn't just told what my research topic was.' Amanda similarly described the many classes 'where I had to read what the professor wanted me to read.' Scholarly digital storytelling, in contrast, taught 'me how to do my own research and dig into primary sources' and allowed her the opportunity to find 'something I was interested in.' She noted that 'it was really a starting point for me to take me into deeper research in my field that I really wanted to do … finding the thing that I could endlessly research and still find interesting.' This was especially meaningful for Amanda because she shared that she had seriously considered leaving graduate school, but decided to stay after this experience. During the course, she reflected in a blog post, 'the most important thing I learned … is that I love history … . I loved doing my own research on something I was personally interested in.'

For other students, such as Mark, digital storytelling led him to rethink his research agenda. He had written many papers on college access, but creating a digital story helped him see it in a new way:

> There's an interesting conflict that this course actually brought out in my own research question … . if you look at my story, it's not really about college access at all … . it's more about helping explain something that's in a black box … there's a question that needed to be answered before that and this forced me to do that.

Creating a scholarly digital story enhances traditional research skills and at the same time shapes student perceptions of disciplinary methods through an emphasis on independent

thinking. Karen discussed how she came to appreciate the process as 'a form of research in and of itself' including a 'new messy form of thinking.' Chelsea shared that she was able to integrate a 'feminist framework in a meaningful way … [through] a collaborative storytelling experience in which I can offer as much agency as possible to my participants.'

Several students shared that after the course ended, they incorporated scholarly digital storytelling into academic portfolios, conference presentations, and subsequent courses while others left the class with renewed enthusiasm for academic research. Chelsea remarked that the experience helped her 'come back to life in terms of, sometimes you just get into academic ruts and the digital storytelling class gave me a new way to approach it.' Taylor wrote in a final class reflection, 'Thinking about history in this way makes it fun and stretches the boundaries of conventionality that I often feel weighing me down when doing more "traditional" history.' This comment is inherently about the blending of inquiry-based learning with multimodal research and knowledge production. The autonomy to define their own research while exploring and experimenting with new forms of scholarly communication allowed students to appreciate the possibilities inherent in scholarly work.

Flexible thinking

Knowledge production requires flexible thinking through the practical experience of doing – of creating, revising, rethinking, and revising yet again. This is inherent in the process of scholarly digital storytelling and is taught explicitly throughout the semester. Students frequently recalled the small assignments that helped them build the skills necessary to complete the final project, such as a one-minute video introducing the digital story concept early in the semester. Students received extensive feedback from classmates and the teacher and were required to submit a revised video the following week. This pattern repeated throughout the course and students developed an awareness of, and appreciation for, flexible, iterative thinking by regularly revising assignments and building week by week toward the final project.

Bailey recalled the process of figuring out 'how you're going to translate that conceptualization into an actual script, a storyboard, and all those different parts … actually, it's a lot of investment.' Chelsea accepted this as central to the project and shared that 'There's a lot of pieces, it builds … . And you're never really done, so don't let that bother you.' While a traditional research paper includes multiple steps, the multimodal nature of scholarly digital story production engages students in iterative development in new ways and Chelsea's comment demonstrates growing awareness of the importance of flexible thinking as a valuable part of the intellectual process.

This experience shaped the topic development for Ashley who recalled, 'You come up with an initial storyboard thinking this is the direction my video will go in and this is how I'll tell my story. But of course that's not how it works.' She appreciated her growing capacity for academic research, 'something that's always a work in progress or always something to adjust I think keeps you focused more as far as making the final decisions and editing and deciding what to include and what to exclude.' Ken summarized this concept in a project update:

> So in some ways this is a "two steps forward, one step back" post except that I don't think the creative process is that simple. Once you stumble upon a good idea, you can't bemoan the wandering path you took to arrive at that point. Every step of the journey was necessary.

Editing, a central task in digital storytelling, provides additional opportunities to engage in flexible thinking. Students often initially experienced this as frustrating, but learned to appreciate the process as both scholarly and productive. Tracy recalled:

> It took me a while to come up with that opening. I tried and rejected several others … that was not the second or probably even the third or maybe even the fourth try at an opening that would be effective … . But once the final idea came along, I did recognize that, "Okay, this is it."

Adam reflected on the hard decisions he made throughout multiple versions of his digital story:

> The first one was a mess because of some of the technical problems that I was having … . The second draft I remember was just way too long. And that's when I realized, "OK. Yeah, I have all this great information, but this isn't like writing a paper." You've got to cut it down and only give the most important or the most powerful, most compelling pieces of evidence to get your point across. So I had to go back and really cut a lot out and do some rearranging.

Adam had to rethink not only what evidence to include, but how to effectively communicate an argument based on evidence. He had to rethink his approach, his decisions about what content to include, and how to best convey his work.

Through this experience, students developed a capacity for flexible thinking that they could continue to utilize in the future (Willison et al., 2017). Chad, for example, continued to use storyboarding to develop online courses because he found the process productive and meaningful. Ashley commented that 'even though I'm not making a video for my thesis paper, I keep writing out a tentative storyboard that I keep editing as I go along.' The focus on ongoing editing shaped her process and encouraged her to remain flexible as she analyzed sources and crafted her emerging argument. These examples demonstrate the ways in which digital storytelling, including structured aspects of the process such as storyboarding, required students to reimagine and rethink their approach to scholarly research and knowledge production. They also indicated ways in which they applied these skills and strategies in other contexts beyond the course.

Multimodal research

Scholarly digital storytelling students engaged with traditional academic research practices according to their disciplines while they learned the craft of digital knowledge production, moving beyond the 'hegemony of text-based knowledge in academic scholarship' (Literat et al., 2018, p. 565). Students filmed interviews and analyzed a diverse array of sources, including video and audio. They incorporated data into graphics, data visualizations, and maps and puzzled over how to tell their stories visually, communicate their arguments concisely and effectively, and maintain viewer interest through pacing and music. These examples indicate the ways in which digital storytelling promotes awareness of digital research, digital knowledge production, and the capacity to

engage with multimodal sources in new ways. These skills are explicitly taught through assignments, such as telling a story with images, and through ongoing formative feedback focused on individual project development.

Taylor analyzed a fictional movie depicting a historical event while asking larger questions about filmic representations of the past. Taylor shared that they took the opportunity to 'really dig into this project,' including taking leave from work and traveling to another state to access analog audio recordings of eye witness interviews stored on cassette tapes. Taylor recalled that 'I spent the whole day with some headphones on and just listened. And that was a long, long day, but it was great … it's one of my favorite memories of my degree program.' Taylor digitized the audio recordings in order to integrate voices from the past into the digital story in an authentic and impactful way.

Other students were challenged by an abundance of resources. Amanda remembered feeling initially 'like I was drinking from a firehose trying to narrow down my topic' because she wanted to shed light on a familiar subject in a new way. 'There are so few suffragists that we recognize now, but so many women worked so hard to get the vote. I think it is important to tell all of their stories.' She selected a magazine illustrator as her core focus due to the visual nature of the digital story knowing that she could engage the audience in an intimate and immediate way through the drawings.

Adam recounted that he learned to take advantage of the multimedia affordances of his content in a way that was 'completely different than writing a history paper. I brought in music from the 1800s. I had different people read newspaper articles and speeches so you have an element of real people and not just words on a page.' This was a common experience as students learned to think critically about the multimodal components of their projects. Roberto learned through his project on the 'Dirty War' in Argentina that 'the quietest voices can often be heard the most.' A slow pan across a wall of faces, photographs of the *desaparecidos* or disappeared, conveyed the magnitude of the violence and its lasting impact on families and society more effectively than narration or text on a screen.

These kinds of experiences also influenced student thinking about digital scholarship. Students came to recognize that scholarly digital storytelling required significant intellectual effort. Lauren recalled:

> [I] got to a point where I was in the library one day for eight hours working on this and I was like, "man, I would have knocked a paper out in three hours." And I thought this was going to be an easier alternative and it really wasn't.

Students regularly made these kinds of comparisons when reflecting on the course in contrast with previous higher education experiences. In retrospect, Morgan said, 'I think I learned the history better doing this project than writing a history paper.' Bailey stated that the digital storytelling format demanded more accountability, 'if you're writing, you can just add a fluff paragraph and at least you're hitting the word count.' With the digital story, however, every word, every image, and every sound mattered.

While identifying, accessing, and using sources are essential research skills that can be taught in traditional disciplinary contexts, these examples demonstrate ways in which the multimodal nature of the technology-enhanced assessment shaped the learning

experience for students. They engaged with visual and aural content essential to digital knowledge production and it shaped their research skill development in important ways.

Limitations

While students in this study represent a range of disciplines and academic programs, it originates from courses taught by one faculty member at one institution. This teacher is experienced in teaching digital storytelling and accepts the risks, such as technical challenges, involved in designing a course around an immersive, technology-enhanced assessment. This course is also designed as a full-semester course which may not be feasible for all disciplines or teaching and learning contexts. Schrum, the teacher, conducted the interviews which may have introduced the potential for social desirability bias (Larson, 2019). Miller (2012) has found minimal impact of social desirability bias in student survey responses related to student engagement and learning (p. 45), but this still may be a factor in individual, in-person interviews with the teacher. In addition, Schrum was one of the researchers involved in coding and analyzing the data so may be influenced by knowledge about the course or individual student work beyond the data collected for this study. We attempted to mitigate this by having the second author independently code and analyze data followed by collaborative review.

Discussion

Attention to student research skills has led to an increased focus on intentional skill development throughout the higher education curriculum (Willison, 2018). Willison (2018) and other supporters of open inquiry have advocated for rethinking the role students play in their own learning process as well as redesigning assessments to meet current and future needs inside and outside of academia (Gyuris, 2018; Rodríguez et al., 2019). This increasingly requires attention to digital as well as traditional research skills (Carretero et al., 2019; Literat et al., 2018). This research offers an approach to combining these pedagogical approaches and imperatives through technology-enhanced assessment.

In alignment with the RSD framework and digital literacy skill development, findings indicate that scholarly digital storytelling has the potential to promote student development of research skills, including autonomy, flexible thinking, and multimodal research, across humanities and social science disciplines. Morgan and Amanda, for example, defined their own research topics for the first time and found the experience rewarding and engaging. Other students, such as Taylor, appreciated the opportunity to rethink traditional disciplinary boundaries. Students such as Ashley and Ken learned that scholarly thinking is iterative. Ultimately, authentic digital projects empowered students to evaluate and produce digital content and communicate their scholarship effectively in a new medium (Literat et al., 2018; Willison, 2018). The scaffolded building blocks, such as storyboards and treatments, required students to focus on the pieces as well as the whole and to rethink their evidence, argument, and presentation multiple times. They experienced scholarly research as non-linear and articulated the ways in which this experience differed from traditional research papers.

Finally, in creating their digital stories, students engaged in multimodal research, learning to locate, create, and analyze a range of texts, objects, graphics, audio, and video – important skills that enabled the production and sharing of knowledge in new formats (Literat et al., 2018). Taylor tracked down and digitized cassette tape recordings to bring voices from the past to new audiences while Amanda shaped and refined her topic around visual evidence to maximize the affordances of digital storytelling. They learned the importance of each element of a digital story by creating, sharing, receiving feedback, and revising and came to appreciate, as Bailey stated, that digital knowledge production required their full attention. A scholarly digital story is not something you can fill with 'fluff' or 'knock out' in a few hours.

While these students had previously learned traditional research skills grounded in disciplinary thinking, scholarly digital storytelling engaged them in digital research and challenged their habitual and familiar approach to knowledge production. Students took an active approach toward problem-solving and demonstrated resilience and willingness to seek alternative solutions (Iordache et al., 2017; Willison et al., 2017) even when the work was frustrating or required more effort than anticipated, as noted by Lauren and Tracy. Students, as demonstrated through interviews, course work, and the digital stories themselves, came to appreciate the value of academic research within their disciplinary contexts as well as the role of persistence and flexible thinking in their own practice. The act of producing digital stories grounded in scholarship led some students, such as Mark, to rethink the nature of their research and others to reimagine future research and how it would be received by an intended audience (Schrum, 2021; Willison, 2018).

Implementation of scholarly digital storytelling assignments requires effort. These skills are intentionally integrated into the course structure and assignments and are explicitly discussed in class and through feedback. In programs with formal coursework and assessments, this could include rethinking learning objectives, assignments, and pacing as well as introducing new, tangible ways to value digital knowledge production (Literat et al., 2018). In programs with less formal coursework, this could be integrated into independent research along with timely feedback, scaffolding, and digital skill support. Scholarly digital storytelling can also be incorporated through smaller assignments, such as research overviews or digital conference presentations. There are also digital storytelling tools, such as Canva and Adobe Spark, that require little technical experience and allow students to share their findings in new modes and formats, but even these require teachers to think critically about learning goals and to dedicate time to restructuring traditional assessments. This research, however, suggests that these efforts can positively support student development of valuable research skills.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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